CURRICULUM VITAE
Xinjian Zhao

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EDUCATION AND ACADEMIC VISITS

The Chinese University of Hong Kong, Shenzhen ***Sep. 2023 – Oct. 2027 (expected)***
Fourth-year Ph.D. student in Computer Science. Advisor: Prof. Tianshu Yu.
Research interests: representation learning, generative modeling, and reasoning, with applications to graph learning and AI for science. Current topics include chemical reasoning, multimodal molecular embeddings, retrosynthesis, and graph self-supervised learning.

NLPR, Institute of Automation, Chinese Academy of Sciences ***Oct. 2024 – Jul. 2025***
Visiting Ph.D. student at the State Key Laboratory of Multimodal AI Systems. Host: Prof. Shu Wu.

City University of Hong Kong ***Sep. 2021 – Oct. 2022***
M.S. in Data Science. Supervisor: Prof. Ruocheng Guo.

Shandong University ***Sep. 2017 – Jun. 2021***
B.S. in Computer Science. Supervisor: Prof. Xuemeng Song.

PUBLICATIONS

* denotes equal contribution; underlined names indicate mentees.

First- and Co-first-authored Papers

[1] **Xinjian Zhao**, Yaoyao Xu, Xuemin Chen, Xiaozhuang Song, Tianshu Yu. "Latent JEPA: Abstract Future Prediction for Latent Reasoning in Chemistry." ***Preprint***, 2026. [PDF]

[2] **Xinjian Zhao**, Xiangru Jian, Yaoyao Xu, Xiaozhuang Song, Wei Pang, Lei Bai, Tianshu Yu. "MolEmb: Multimodal Large Language Models Can Be Strong Molecular Embedding Models." ***Working Paper — ICML FM4LS Workshop***, 2026. [PDF]

[3] **Xinjian Zhao**, Wei Pang, Zhongkai Xue, Xiangru Jian, Lei Zhang, Yaoyao Xu, Xiaozhuang Song, Shu Wu, Tianshu Yu. "The Underappreciated Power of Vision Models for Graph Structural Understanding." ***NeurIPS***, 2025. [PDF]

[4] **Xinjian Zhao**, Wei Pang, Zhixuan Yu, Xiangru Jian, Xiaozhuang Song, Yaoyao Xu, Zhongkai Xue, Dingshuo Chen, Shu Wu, Philip Torr, Tianshu Yu. "When Vision Meets Graphs: A Survey on Graph Reasoning and Learning." ***IJCAI***, 2026. [PDF]

[5] **Xinjian Zhao**, Chaolong Ying, Yaoyao Xu, Tianshu Yu. "Graph Learning with Distributional Edge Layouts." ***SIGKDD***, 2025. [PDF]

[6] **Xinjian Zhao**, Liang Zhang, Yang Liu, Ruocheng Guo, Xiangyu Zhao. "Adversarial Curriculum Graph Contrastive Learning with Pair-wise Augmentation." ***arXiv***, 2024. [PDF]

[7] Hao Xu*, Xiangru Jian*, **Xinjian Zhao***, Wei Pang*, Chao Zhang, Suyuchen Wang, Qixin Zhang, Zhengyuan Dong, Joao Monteiro, Bang Liu, Qiuzhuang Sun, Tianshu Yu. "GraphOmni: A Comprehensive and Extensible Benchmark Framework for Large Language Models on Graph-theoretic Tasks." ***ICLR***, 2026. [PDF]

[8] Xiangru Jian*, **Xinjian Zhao***, Wei Pang*, Chaolong Ying, Yimu Wang, Yaoyao Xu, Tianshu Yu. "Rethinking Spectral Augmentation for Contrast-based Graph Self-Supervised Learning." ***TMLR***, 2025. [PDF]

[9] Maolin Wang*, **Xinjian Zhao***, Wanyu Wang, Sheng Zhang, Jiansheng Li, Bowen Yu, Binhao Wang, Shucheng Zhou, Dawei Yin, Qing Li, Ruocheng Guo, Xiangyu Zhao. "Embedding in Recommender Systems: A Survey." ***ACM TOIS***, 2026. [PDF]

[10] Yaoyao Xu*, **Xinjian Zhao***, Xiaozhuang Song, Benyou Wang, Tianshu Yu. "Boosting Protein Language Models with Negative Sample Mining." ***ECML-PKDD***, 2024. [PDF]

Selected Co-authored Papers

[11] Yaoyao Xu, **Xinjian Zhao**, Xiaozhuang Song, Xuemin Chen, Tianshu Yu. "ChemOPD: Affinity-Guided Multi-Teacher On-Policy Distillation for Chemical Reasoning." ***Preprint***, 2026.

[12] Yaoyao Xu, **Xinjian Zhao**, Lei Bai, Tianshu Yu. "Closed-Loop Bayesian Molecular Inverse Design with Semantic LLM Surrogates." ***Preprint***, 2026.

[13] Zhixuan Yu, **Xinjian Zhao**, Tianshu Yu. "TAPER: Task-Adaptive Pairwise Energy Regularization for Graph Neural Networks." ***WAIC Academic***, 2026, Shanghai, China.

[14] Chaolong Ying, **Xinjian Zhao**, Tianshu Yu. "Boosting Graph Pooling with Persistent Homology." ***NeurIPS***, 2024.

[15] Xiaozhuang Song, **Xinjian Zhao**, Lei Bai, Tianshu Yu. "Geometric Grounding Implicitly Unlocks Structural Reasoning in Chemical Language Models." ***Preprint***, 2026.

[16] Xuemin Chen, Xiaozhuang Song, **Xinjian Zhao**, Yaoyao Xu, Tianshu Yu. "Coverage Before Control: Route-Instruction Grounding and Steering for Controllable Retrosynthesis." ***Preprint***, 2026.

[17] Xiaozhuang Song, Xuanhao Pan, **Xinjian Zhao**, Hangting Ye, Shufei Zhang, Jian Tang, Tianshu Yu. "AOT*: Efficient Synthesis Planning via LLM-Empowered AND-OR Tree Search." ***Findings of ACL***, 2026.

[18] Xiangru Jian, Hao Xu, Wei Pang, **Xinjian Zhao**, Chengyu Tao, Qixin Zhang, Xikun Zhang, Chao Zhang, Guanzhi Deng, Alex Xue, Juan Du, Tianshu Yu, Garth Tarr, Linqi Song, Qiuzhuang Sun, Dacheng Tao. "FORGE: Fine-grained Multimodal Evaluation for Manufacturing Scenarios." ***Findings of EMNLP***, 2026.

[19] Wei Pang, Xiangru Jian, Hehan Li, Zhixuan Yu, Jingxi Xue, Jinyang Li, Zhengyuan Dong, **Xinjian Zhao**, Hao Xu, Chao Zhang, Reynold Cheng, M. Tamer Özsu, Tianshu Yu. "TRL-Bench: Standardizing Cross-Paradigm Representation-Level Evaluation of Tabular Encoders." ***Preprint***, 2026.

ONGOING RESEARCH

- **Graph Self-Supervised Learning.** I investigate how interactions between graph populations shape learned representations. Using a manifold-based perspective, I study representation transformations that separate entangled populations into orthogonal subspaces while preserving cohesion within each population. This work aims to provide principled learning objectives beyond augmentation heuristics.

AWARDS

- NeurIPS 2025 Scholar Award, 2025
- Duan Yongping Scholarship, 2025
- National Second Prize, Chinese Mathematics Competitions for College Students, 2020
- Student of the Year Award, Shandong University, 2019
- Scholarship of Academic Excellence, Shandong University, 2018

PROFESSIONAL SERVICE

Journal reviewer: IEEE TPAMI, JMLR, ACM TOIS, IEEE TCSVT.
Conference and workshop reviewer: NeurIPS, ICLR, KDD, KDD AI4S, ECCV, AAAI, EMNLP.